

VINEET KUMAR

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Summary

M.S. Computer Software Engineering student at **San José State University**, available **June 2027**, with **3+ years** building production software at **Microsoft** and **ION Trading**. Builds **AI systems** and **large-scale infrastructure** on **distributed systems**, with work in **workload placement**, **multi-agent LLMs**, **RAG**, and **voice AI**. Co-founder of **TrueStar**.

Education

Master of Science, Computer Software Engineering | **San Jose State University** **August 2025 to May 2027**
Distributed Systems | Machine Learning & Data Mining | Operating Systems | Enterprise Software
Bachelor of Technology, Computer Science | **Vellore Institute of Technology** **July 2018 to June 2022**
GPA 3.87 | Data Structures & Algorithms | Computer Architecture | Information Security

Experience

Microsoft | *Large-scale data-ingestion infrastructure powering Microsoft Sentinel's data lake*
Software Engineer Intern **May 2026 to August 2026**

- Built an **intelligent workload-placement engine** in **C#** and **.NET** that routes about **82K shared-tenant pipelines** onto right-sized **Kubernetes** compute, engineered to raise fleet **CPU utilization** about **5x (11% to 59%)** and cut compute cost.
- Designed a two-step, **config-driven placement algorithm** with a network-boundary (**VNet**) **match guard** that structurally prevents mis-routing across **isolated networks**, shipped behind **feature flags** for zero-downtime, reversible per-route rollout.
- Analyzed **603K production runs** across 30 days in **Kusto (ADX)** on the **Azure Data Lake** platform to correct a workload-classification assumption, hardening it with **22 integration tests** and a **classifier suite**.

ION Trading | *Trade-processing systems for global financial institutions*
Software Development Engineer **January 2022 to August 2025**

- Engineered a **trade-processing framework** with **Apache Camel** and **Java**, integrating **Kafka** and **ActiveMQ** for **ExxonMobil** to enable real-time, **large-scale** trade-data exchange and lower latency.
- Enhanced a **commodity-trading module** with **Java Native Interface** and **C++** to automate bulk trade scheduling, cutting UAT time from **4 weeks to 30 minutes**.

Projects

TrueStar | *Co-Founder, AI-Native Research Platform* truestar.tech

- Architected an **AI-native platform** of **4 microservices** including **TARS**, a **multi-agent LLM** engine with a **hallucination-audit** layer, plus a **RAG service (FastAPI, Pinecone)** with **embeddings** and a **Cohere cross-encoder reranker**.
- Shipped a real-time **voice AI** interviewer on **LiveKit WebRTC**, integrating **Deepgram STT**, **GPT-4o**, and **Fish Audio TTS** at about a **700ms** round-trip, tuned for low-latency **inference** with per-expert **namespace isolation**.

Self-Healing AI Ops Mesh | *Distributed Systems / gRPC / Observability*

- Built a self-healing **gRPC** service mesh of **16 containers** and **8 microservices** that detects failures via **2-of-3 statistical consensus** and auto-remediates through an **LLM reasoning layer**, instrumented for **reliability** with **OpenTelemetry**, **Jaeger**, and **Prometheus**.

Distributed Fire Query System | *C++ / Distributed Computing / gRPC*

- Implemented a distributed **C++** query engine over **1.17M records** using **gRPC** and **Protocol Buffers** with **6 processes**, **Bully leader election**, health checks, and fault-tolerant **work redistribution**, hitting a **2.77x OpenMP speedup**.

Achievements & Leadership

- Won **2nd Place**, Code Run Seek Hackathon (IEEE IAS & WIE) at VIT and **Beta-rank Microsoft Learn Student Ambassador**.
- Authored a **Data Structures & Algorithms** repository used by **250+ students** at VIT.

Skills

Programming Languages: Python, C++, Java, JavaScript, C, C#, TypeScript, SQL, Bash

AI / ML: ADK, A2A, AI agents, LLMs, RAG, machine learning, inference, embeddings, vector search, Pinecone, FAISS, Azure OpenAI
Infrastructure / Distributed: distributed computing, system design, microservices, gRPC, Kafka, Kubernetes, observability

Databases / Storage: PostgreSQL, Redis, MySQL, MongoDB, Cassandra, Oracle

CS Fundamentals: data structures & algorithms, operating systems, networking, security, concurrency

Containers / Cloud / Tooling: Docker, Azure (Data Lake, ADX), AWS (EC2, S3, RDS, EKS), Git, CI/CD, unit testing